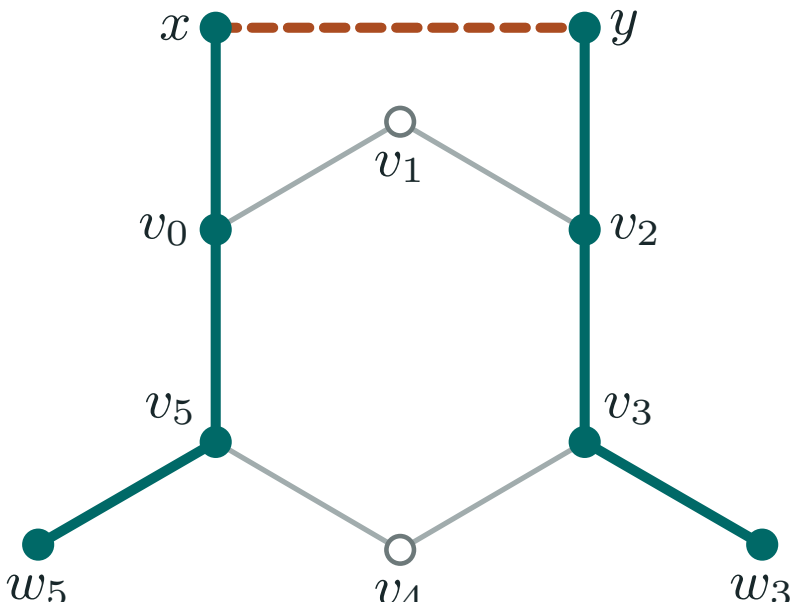
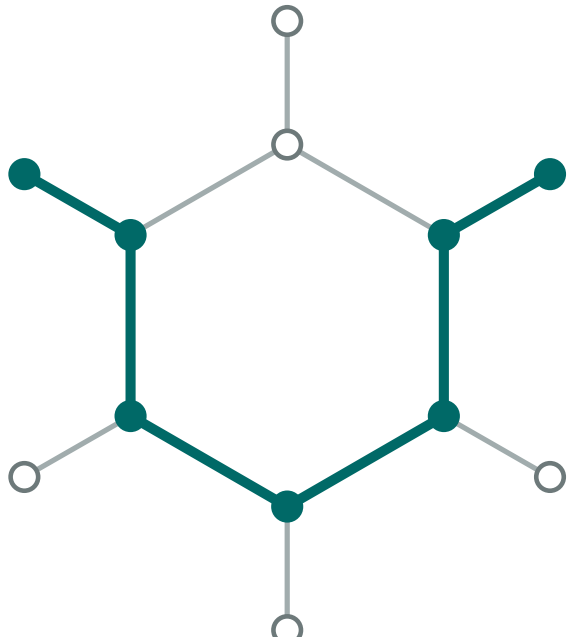
(a) A forbidden cross edge



The dashed edge xy completes an induced P_8.

(b) Equality in $S(G) \leq 2n$



$r = 4, \quad p = 7, \quad S = 24 = 2n.$